MSBA 402 – Fall 2026

Intro to SQL and Basic Data Management

Lecture 1

September 28, 2026

Prof. Ryan Rosario

Anderson School of Management – Business Analytics

University of California, Los Angeles



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Outline

- Introduction and Fun Stuff
  - A Bit About Me
  - About Your TA
- Syllabus / Logistics
- Introduction to Databases
- SQL and Data Management Diagnostic and Growth Assessment

Welcome to Fall Quarter!



Welcome to Fall Quarter!



Yes, California has Seasons!



Yes, California has Seasons!



Getting to Know You

Now it's time for the roast...



This Section

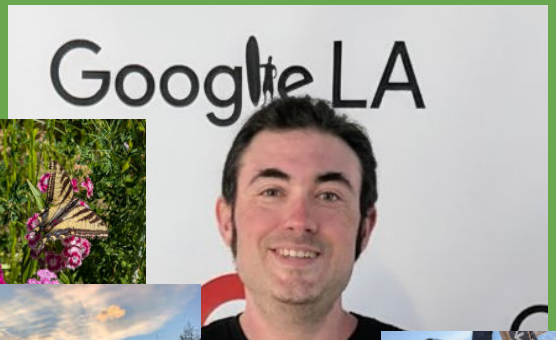
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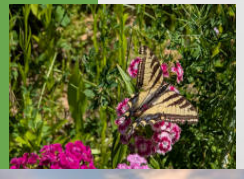
My Slide

Click Here

Ryan Rosario















- 🏠 **Hometown:** Thousand Oaks, CA, USA
- 🏠 **Current home:** Mammoth Lakes, CA USA
- 🎓 **Educational background:**
  Statistics, Ph.D., Computer Science, M.S., Mathematics, B.S. All at UCLA
- 💼 **If I weren't in tech I would work in:** Educational technology and assessment, aviation, full time teaching, or running a ranch (like this place)

Accomplishments/Jobs:

- Current Software Engineer at Google (Google Kubernetes Engine)
- Previous Data Scientist at Google (Search)
- Previously at Amazon and Facebook/Meta and Educational Testing Service (ETS)

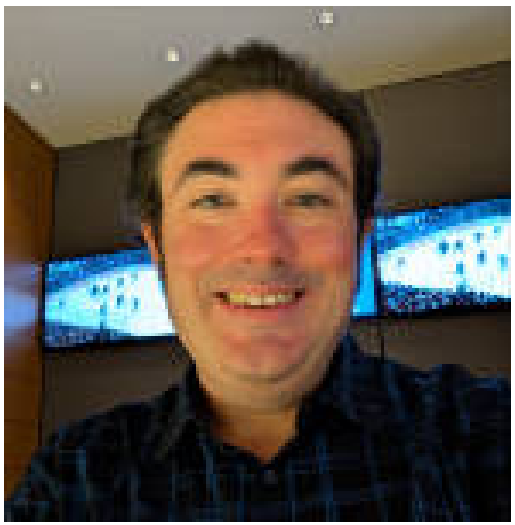
Hobbies:

- Mountain biking! I am a ◆ rider. I also coach the local mountain biking team (5th/6th grade boys). This summer I also mountain biked in Scotland and Lake Tahoe.
- Gardening: lots and lots of flowers and berries
- Aviation and airports are two of my special interests.
- Aviation and airports are two of my special interests.

Fun Facts

- My granduncle, Harold Arlen, wrote the music for *The Wizard of Oz*.
- I like anything tropical, anything with strawberry and anything with peanut butter.
- I just visited Scotland for the 5th time an stayed in a castle!

whoami



UCLA

Ph.D., *Statistics*
M.S., *Computer Science*
B.S., *Mathematics of Computation*

I've been at UCLA forever...

My Contact Information

My email is `rrosario@cs.ucla.edu` but please use it for discussing a **personal matter** with me. My Anderson email (`ryan.rosario@anderson.ucla.edu`) is forwarded to my CS email address.

My office hours:

- Monday, 6:30–7:30pm in C Building Lobby
- Wednesday, 7:00–8:00pm on Zoom

Any changes will be communicated to the class in writing.

If there is a personal matter that is better discussed in person, we can schedule an **appointment**, likely on Zoom.

Professional Interests

Academic / Research Interests



Machine Learning / AI



Natural Language Processing



Database Systems



Distributed Systems

Analytics Interests



Educational Measurement/Technology



Operations Research

Professional Interests



These all either require a lot of data, or help serve data, powering insights.

I am From Industry

Some places I've worked:













Now: Software Engineer at Google



I currently work at Google as a Software Engineer working on productionizing AI and ML workloads on Google Kubernetes Engine.



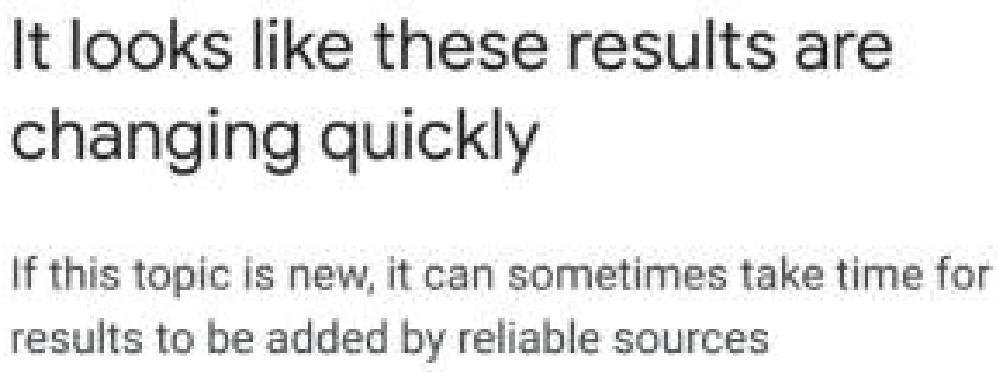
Previously: Data Scientist at Google in Search



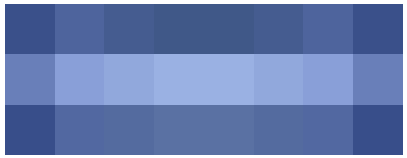
- Develop metrics, experiments and testing methodology to make sure that new search ranking algorithms or features positive affect user experience
- Support engineering launches by checking important user satisfaction metrics.
- Develop and rigorously test new metrics and statistical methodology for measuring user experience across very complex systems with many interactions.

Previously: Data Scientist at Google in Search





What I Did at Facebook: Machine Learning Engineer



- Developed the company-wide sentiment analysis system.
- Developed a system for topic identification and topic summarization.
- Worked on automated tagging of IT tickets.
- Worked with HR's People Analytics team using classical statistics to analyze inequities in performance review text and leveling by demographics.

This Section

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  - A Bit About Me
  - About Your TA
- Syllabus / Logistics
- Introduction to Databases
- SQL and Data Management Diagnostic and Growth Assessment

Your TA

Your TA for this class is **Fauzan Budi Prasetya**. He is a master's student in Data Science in Health.



Email:
fauzan.prasetya@ucla.edu

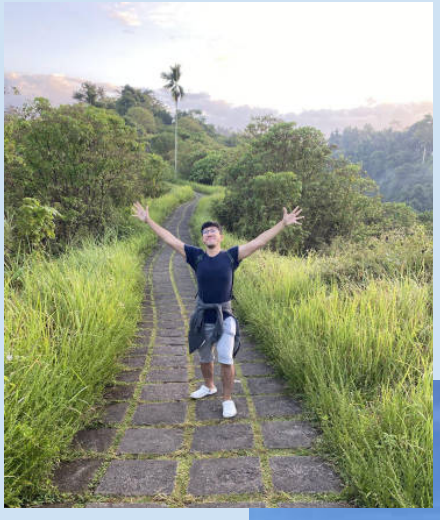
Office Hours:
Thursday,
11:00am–12:00pm on Zoom

Discussion:
Wednesday 12:00–1:00pm,
Collins A301

Fauzan's Slide

Click Here

Fauzan Budi Prasetya





- **Home city, state, country:** Bekasi, Indonesia
- **Graduate major:** Master's Data Science in Health, 2nd years
- **Undergraduate University:** University of Indonesia
- **Undergraduate major(s):** Public Health and Biostatistics

Work/education experience:

- Data Lead @ Segari
- Senior Data Analyst @ Paper.ID
- BI Data Analyst @ Tokopedia

Hobbies:

- Going to the gym
- Just got into hiking and cycling
- Enjoy playing RPGs and watching anime

Fun Facts:

- My name comes from Arabic and Sanskrit
- Fauzan is pronounced as faw-zuhn
- Supported by a Fulbright scholarship
- Planning to try martial arts classes at UCLA Rec

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Syllabus



BUSINESS ANALYTICS 402

Intro to SQL and Basic Data Management

Fall 2026

Prof. Ryan Rosario

Instructor Contact Information

Email: rosario@ucla.edu
Bio Website: https://ryanrosario.phd
Teaching Site: https://pub.cs.ucla.edu/~rrosario
Office Hours: Monday, 6:30–7:30pm in C Building Lobby
Wednesday, 7:00–8:00pm on Zoom
Day/time/modality/location is subject to change.

Instructor Background

Ph.D., *Statistics*, UCLA.
M.S., *Computer Science*, UCLA.
B.S. *Mathematics of Computation, Statistics*, UCLA.

I am from industry. I am originally from Thousand Oaks, CA – born and raised in Southern California and now live just outside Mammoth Lakes, CA in the Eastern Sierra.

Teaching Assistant Contact Information

Name: Fname Bull Pmartyn
Email: fname.pmartyn@ucla.edu
Office Hours: Thursday, 11:00am–12:00pm on Zoom
Discussion: Wednesday 12:00–1:00pm, C-office A301

About MSBA 402

This course teaches you to manage and analyze data in relational database management systems (RDBMS) using SQL: data definition, data manipulation, data control, and analytics. It covers roughly the first four weeks of Computer Science 143 (Databases), another class I teach, without the engineering topics and with additional material specific to business analytics.

This course uses PostgreSQL. Previous offerings of MSBA 402 used MySQL, but PostgreSQL is a better fit for analytics: it supports more sophisticated queries and follows the ANSI SQL standard more closely. Most of what you learn here is standard SQL that works across database systems, and I will point out PostgreSQL-specific syntax when we use it.

The structure of this course has changed. In the past, course content and assessment were based on emulating day-to-day in a corporate environment; however, to get to that corporate environment, students need to pass interviews. Instead, the course is structured around content required to succeed in an interview and at work. Because of this new focus, all assessments are closed-note, closed-book, and closed-computer (often with pencil and paper.

The syllabus is posted on BruinLearn/Canvas.

Please read and review. You are responsible for the policies in it.

Syllabus Quiz

Due to the nature of digital education these days, it is very easy to skip reading the syllabus.

You must take this quiz on BruinLearn. You will receive 50 points towards Participation for completing it. **You can keep resubmitting until you get a score you like.**

Student Survey

You must complete the student survey (click here) to give us a sense of everyone's technical skills, and to tell us about where you've been and where you're going.

You will get 20 points towards Participation for completing it.

Purpose of this Course

- SQL is *extremely* important for business analysts and all careers involving data. It is usually tested in interviews and I've seen many great candidates not pass the SQL part of the interview.
- It is also very easy to make errors in SQL that waste time, waste resources, or crash the entire server.
- This course requires no previous experience with databases or SQL
- My goal is to try to provide value for each student.
- While I want to teach material that is useful for day-to-day work, the more pressing matter is passing interviews.

So, my goal is to give a thorough coverage of SQL and also emphasize some areas that cause trouble.

Piazza



All students have been added to the Piazza discussion forum for this class. Working groups will also be set up.

If you have trouble accessing Piazza, please let us know.

Both lectures are in the same Piazza instance. **Our TA Fauzan will be managing the Piazza.**

Piazza

How we will use Piazza:

1. For course questions, clarifications, confusions, etc. please post a **question** on Piazza.
2. For grading errors, use Gradescope if the assignment is submitted that way. Otherwise, post a **Private Post to Instructors** in Piazza.
3. Please do not send us course material questions via email. *We use Piazza so everyone can benefit.*
4. We *may* use Piazza Live Q&A during lecture (more info in a bit).

GitHub



You may need a GitHub account for this course. You will definitely need one for 405. Please sign up for a GitHub account. If you already have one, you may use it, but you can create a new one if you wish.

In either case, you need to tell us your username here.

Google Docs



Each student has free access to Google Drive.
I will be posting all course materials on Google Drive, and linking to them from BruinLearn.

This also means that if I make changes to the slides, you will always have the most up to date copy!

That said, you will need to routinely log into your UCLA Google Apps account (g.ucla.edu) to access this material.
This is separate from your Anderson email address.

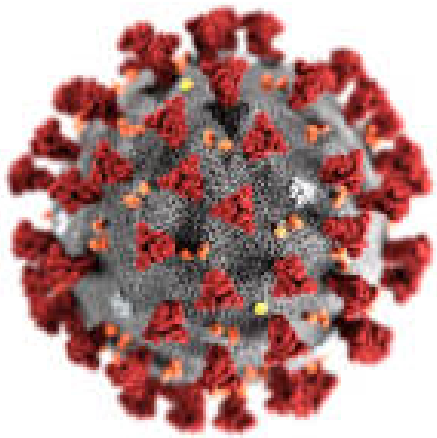
Your Feedback is Important

Student (and instructor) feedback led to several changes in this class:

1. We meet 110 minutes, once per week, over 10 weeks (8 lectures) instead of 3 hours, once per week, over 5 weeks.
2. Extending the length of the couse means more time to practice, more time to absorb the material, and **lower cognitive load.**
3. There will be quizzes, instead of a midterm, to serve as a diagnostic and give students a chance to improve.
4. A formal discussion section with the TA for practice problems, lecture review, interview practice and technical assistance.
5. Several improvements to communication and ways to get help.

I am receptive to feedback about this new model and if there are ways to provide a better experience. If you feel there are topics missing, please let me know.

COVID-19 Protocols: I'm Still Not Gone



Course staff will abide by whatever the current UCLA requirements etc.

If you are sick, please do not come to class. We have plenty of ways for you to catch up.

Your Mental Health Also Matters

You only get one brain, and one life. University life can be challenging. CAPS is there to help (click here).



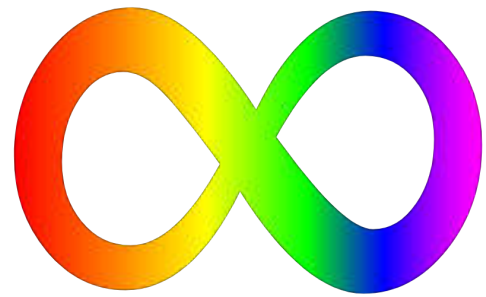


Note that you can also refer yourself to Students in Crisis. Be good to yourselves!

Communication and Empathy

I have recently learned that I am neurodivergent, which can sometimes affect how I communicate.

1. I tend to be very direct—this is to avoid ambiguity in certain situations.
2. My tone of voice and/or facial expressions may not always match the message I intend to convey.
3. I care deeply about your learning and success in this program. Therefore:
   - Please try to assume best intentions in our interactions.
   - I will also strive to do the same for you.
   - These are things I am working on.



Communication and Empathy: Piazza Live Q&A

This year I may use Piazza Live Q& A to try to encourage more engagement. Students choose to not ask questions for a vaiety of reasons. Using techology:

- Reduces concerns about differing technical fluency.
- Reduces concerns about varying levels of English fluency.
- Surfaces common points of confusion through the upvote feature.
- Flags issues to the instructor, such as unclear language or references to unfamiliar technologies.
- Reinforces that there are **no dumb** questions.

I will try not to use English idioms, pop culture references, or sarcasm. It tends to be misinterpreted or simply is not accessible to many students.

My Expectations

My expectations for you:

1. If you have questions, ask them **soon** in some form.
2. If you encounter a technical issue, have a question, or have any other sort of personal matter do not wait to get help. Please research first, then post on Piazza or come to office hours (if soon).
3. Use the course staff, your peers, search and AI to help you (unless instructed not to by the instructor).
4. Assume best intentions – and I will do the same.
5. Behavior in lecture should be professional. Disruptive behavior that distracts other students or the instructor will not be tolerated. Examples include (but are not limited to):
   - Side conversations, including group chats, that are unrelated to course content regardless of volume.
   - Repeated or disruptive laughter unrelated to course content regardless of volume.

My Expectations

You are responsible for all notices and information provided by:

1. The instructor during lecture
2. MyUCLA
3. Piazza
4. BruinLearn

Grades

You are graded according to this schema.

- 15% Attendance & Participation (Attendance via iClicker, course code UIDV)
- 40% In-Class Quizzes (Expect 5 or 6; lowest dropped)
- 45% Final Exam (**12/7 4:00–6:00pm in Anderson Hall G313**)

Grade cutoffs are based on my expectations of the class, not a curve.

Attendance and Participation

You are permitted **one** unjustified absence without penalty to your grade. Justified absences are only for medical, family or unavoidable work-related situations and are at the discretion of the instructor and may require documentation. We will use iClicker to keep records. The tool may change.



You will also be asked to complete certain tasks throughout the quarter, which also counts in this portion of the grade.

Problem Sets

Types of practice on problem sets:

- Types: Writing SQL exercises, using effectively AI, conceptual, LeetCode
- You may use AI unless specifically noted on the problem.
- Certain style is required for SQL queries.
- I try to make this class valuable to all skill levels.

See the syllabus for more information.

Quizzes and Exams

Exams usually consist true/false, multiple choice, short essays and writing queries.

Quizzes will typically be writing SQL queries.

The *exact* format, is to be announced.

No notes, books, or computer. These assessments are pencil and paper.

Regrade Requests

- Submit requests **within 1 week** of grade release (final exam: **within 48 hours**).
- Use **Gradescope** (if applicable), otherwise a **private Piazza post**.
- Limit your request to **150 words**. State which part you believe is graded incorrectly and why.
- Requests must identify a **specific grading error**. Requests to reconsider the rubric, or the size of a correctly applied deduction, will not be considered.
- Queries that produce the correct output but are **conceptually incorrect** will not receive full credit (e.g., they work only by coincidence, omit required conditions, or violate problem constraints).
- A score recorded wrongly in BruinLearn (e.g., not matching Gradescope) can be reported **any time before final grades**, as a private Piazza post.
- Final regrade decisions rest with the instructor.

PostgreSQL

We will use PostgreSQL for this class.



The books I use, as well as my lecture materials, follow the **ANSI SQL standard** which should work across all relational database systems. I will caution the class whenever I teach something that is PostgreSQL specific, or that doesn't work with PostgreSQL.

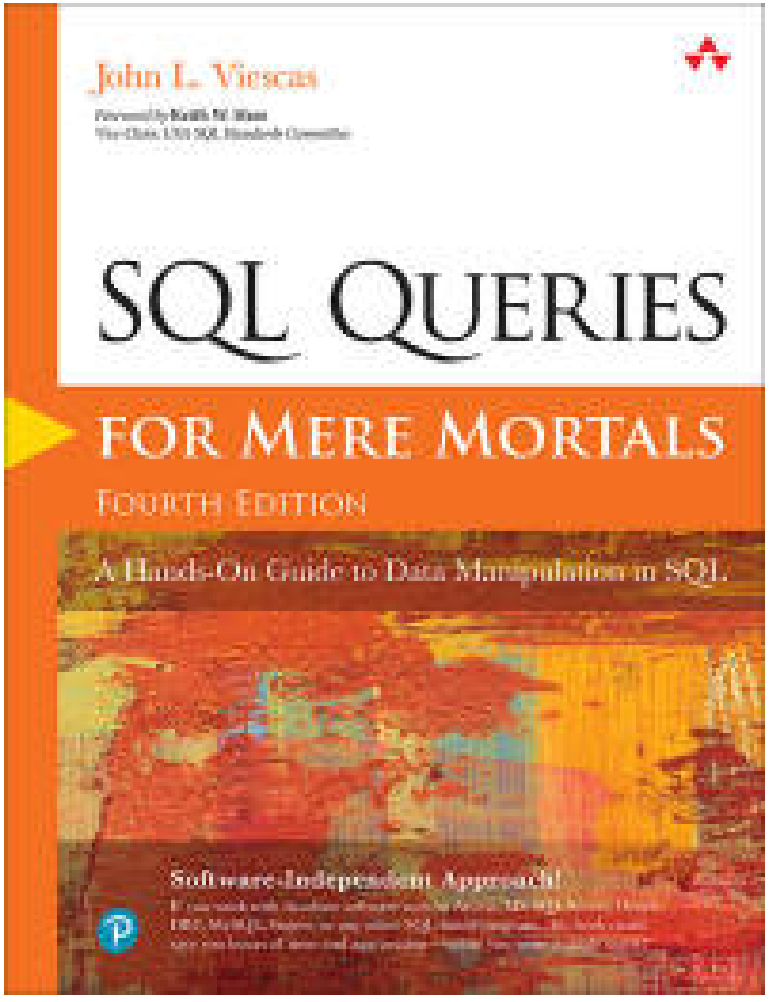
Important!

We focus on the SQL standard so that you can use what you have learned in MSBA 402 in any relational database.



Inspirational Textbook

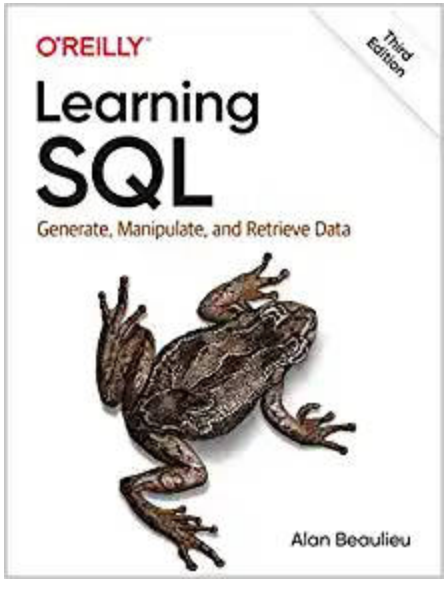
I will be scaffolding my lectures from this book. I may post chapters if I feel they are useful.



SQL Queries for Mere Mortals, Fourth Edition by John Viescas

Reference Textbook

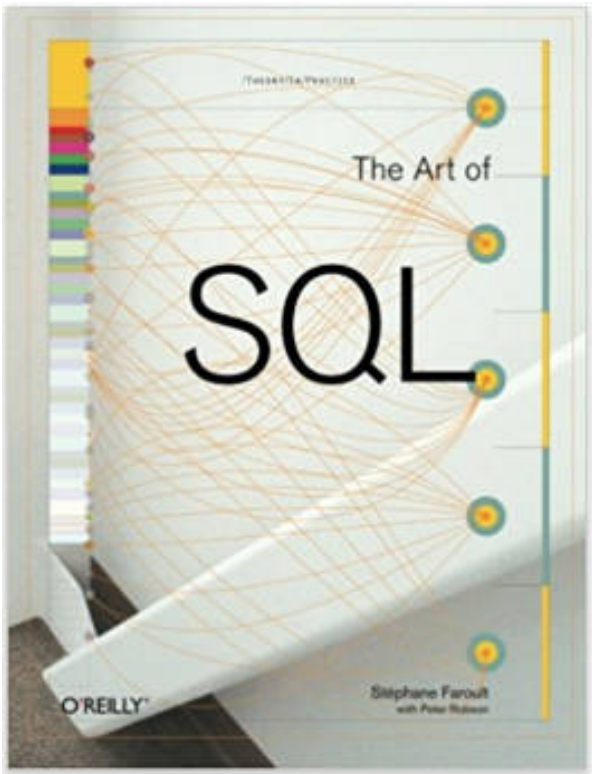
Learning SQL by Beaulieu is straight to the point and is a great reference to have. Everything in this class is covered in the book, but we will go in a different order.



Learning SQL
Alan Beaulieu
This book is free for UCLA students,
but I recommend a print copy.

Other Good Books

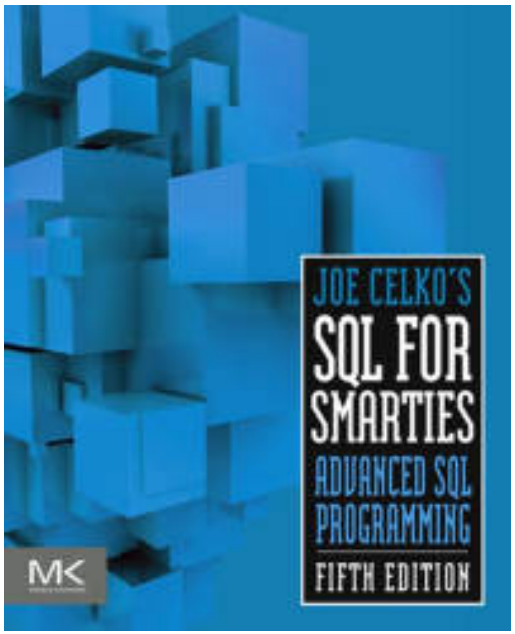
You don't need to buy these and we won't be using them for this course, but they might be worth a look for later on.



The Art of SQL
Stephanie Faroult
Peter Robson

Other Good Books

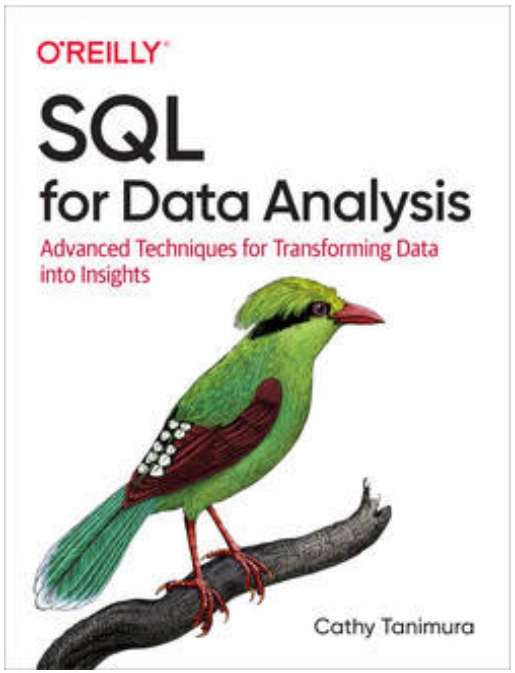
For advanced students:



SQL for Smarties
Joe Celko

Other Good Books

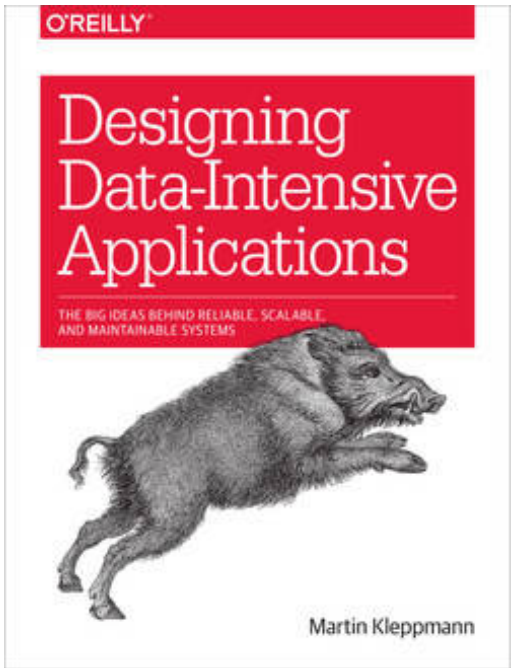
This book discusses how to use SQL for various different types of analysis and may be useful after 402.



SQL for Data Analysis
Cathy Tanimura

Other Good Books

For advanced students, or those interested in data engineering, this book has become an industry standard:



Designing Data-Intensive Applications
Martin Kleppmann

Academic Integrity

We all take academic integrity very seriously. Students that are suspected of violating UCLA's academic integrity policies will be referred to the *Office of the Dean of Students*. Some things to keep in mind (but not limited to):

1. Clarifying requirements of an assignment or sharing approaches **is OK**.
2. Helping a classmate find a bug in their code/work **is OK**.
3. Using code provided in the required textbook, from the instructor, from the TAs, or from reference materials provided in the class **is OK**.
4. Using sources outside of the course staff or materials requires citation.

Academic Integrity

What is **NOT OK**:

1. Using solutions from previous offerings of this or any other class **is NOT OK**.
2. Writing for, or with, another student (aside from your partner) a code fragment that solves any portion of a programming assignment **is NOT OK** unless otherwise indicated by the instructor.
3. Submitting any portion of someone's work, if not from the resources on the previous slide **is NOT OK**.
4. Using AI on any assignment, or part of assignment, where it is indicated that you must not **is NOT OK**.
5. Sharing login information or using another student's login **is NOT OK**

Academic Integrity: AI

AI has revolutionized how people find and interact with information, and has several ramifications for higher education.

- This technology can be very useful for learning as they can be used to create practice problems and provide a solution as you study.
- It is important to understand that these chatbots sometimes return incorrect information or provide output that uses very poor style.
- We want you to use AI efficiency and wisely.

Anderson is supportive of the use of AI tools in classes, it also gives instructors discretion to prohibit its use in their classes.

ChatGPT and similar AI tools are permitted but in some cases the instructor will clearly instruct you not to use it. Please cooperate.

To receive **any** *credit in MSBA 402, you* **must** *sign the academic integrity agreement on BruinLearn.*

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Database Purpose

A database *abstracts away* how the data is stored, maintained and processed. A database:

- Provides a way to create, read, update and delete data without worrying about files and breaking data integrity.
- Provides ONE single location for all data in the database (though the database itself may be distributed over multiple nodes, we don't care).

Schemas

More importantly, the overall *design* of a database is called a *schema*.

The term *schema* is sometimes used interchangeably to describe the logical structure of a **database** or a **relation** (table).

In other texts a *subschema* refers to the design of a particular part of the database, but outside of that text, we usually still call it a *schema*, if it refers to a DB or a table.

Data Model

Data models define how we design databases or interact with data. In particular:

- How do we define data?
- How we encode relationships among data?
- How we impose constraints on data?

In this class, we work with the **relational** model.

The Relational Model

Data is stored as a *relation* (implemented as a **table**). **Rows** represent individual transactions, records, or data points, called *tuples*. **Columns** represent *attributes* common to all rows and have specific *types* and appear in same order across rows.

bldg	**floors**	**wifi_access_pts**	**resident_rats**
Boelter	9	251	0
Engineering VI	5	204	1
SAC	3	98	0
⋮	⋮	⋮	⋮

What is a Relation?

- A *relational database* consists of a collection of *relations* (tables). For the rest of this discussion we will use more conventional terms.
- Each table has zero or more rows. Each row is a transaction, a record, or a data point.
- Each row contains one or more columns. Each column (attribute) has a **type**.
- Each row contains the same columns in the same order.
- A column can contain `NULL` values, which simply means "unknown" or "missing".
- One or more columns form a unique identifier called a **key**.

"Database Languages"

There are many semantic systems for working with databases:

1. Data Definition Language (DDL) *(next time)*
2. Data Query Language (DQL) *(most of the quarter)*
3. Data Manipulation Language (DML) *(next time)*
4. Data Control Language (DCL)
5. Transaction Control Language (TCL)

For the relational model, they all use SQL.

Structured Query Language

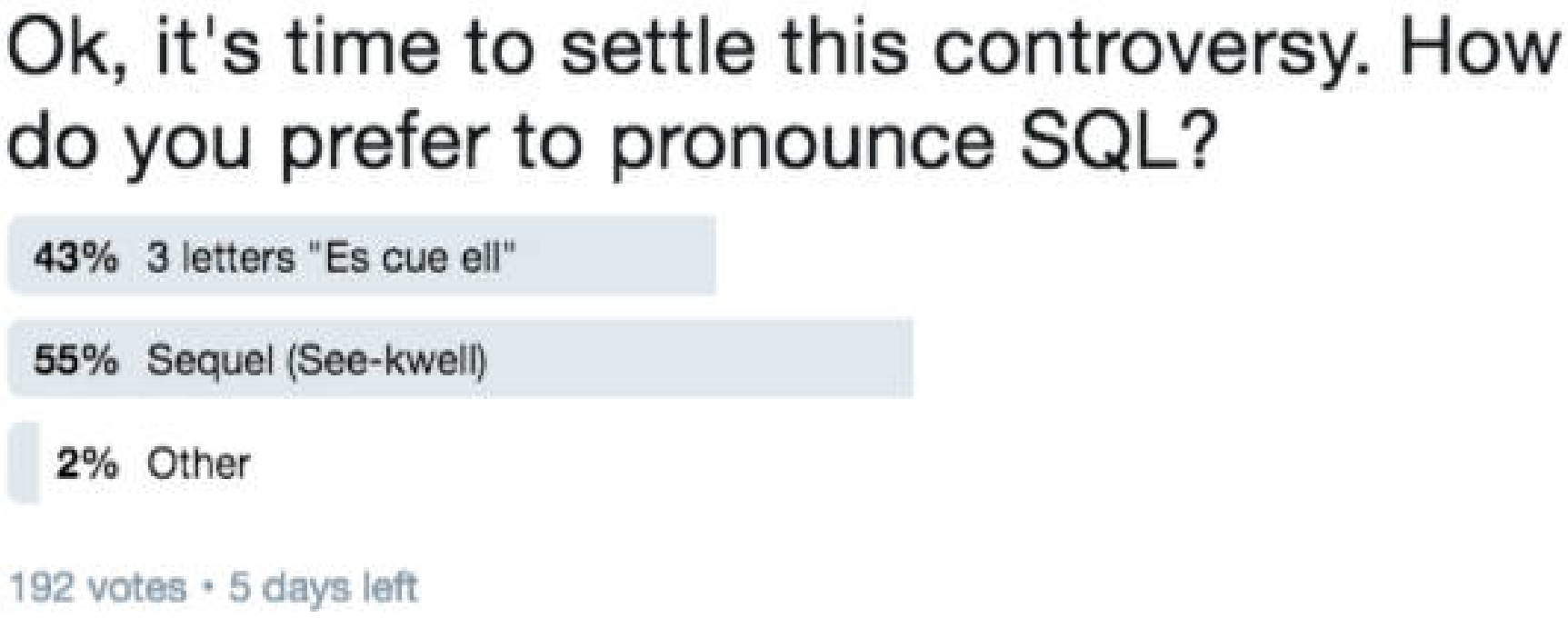
SQL is pronounced as either *Es kyew ell* or *See-kwell* (or "*Sequel*") but I won't judge.



“Database Languages”



You Say Tomayto, I Say Tomahto



Things You Need to Do

Please do the following after class ASAP (including but not limited to):

1. ~~Fill out the student survey for credit.~~
2. Sign the academic integrity agreement.
3. Take the syllabus quiz.
4. Make sure you have access to Piazza and Gradescope.
5. Create a GitHub account and give us the username.

Outline

- Introduction and Fun Stuff
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Materials

A

SQL and Data Management
Diagnostic and Growth Assessment



FORM A
Survey Battery

Form A Survey
DIAGNOSTIC

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SQL and Data Management
Diagnostic and Growth Assessment

FORMS A-G
Complete and Survey Battery

FOR USE WITH Criterion

Student Name _____

UID NUMBER | FORM | BOOKLET # | IMPORTANT

USE #2 (OR HB) PENCIL ONLY

CORRECT

Fill in the circle completely with a heavy, dark mark.

INCORRECT

These marks may not be read properly.

1. If an item has fewer answer choices than this answer sheet provides, leave the extra circles blank.
2. If the test booklet has fewer than 50 items, leave the circles for the unused item numbers blank.
3. Do not make any stray marks on this answer sheet.

Part A — Multiple Choice

DOCULINK ID

SIDE 1

SQL and Data Management Diagnostic and Growth Assessment, First Edition
Forms A-G Complete/Survey Battery; Machine Scorable

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Materials

1. When you receive the answer sheet, **write your name** and fill in the circles with your **UID number**. And then stop.
2. When you receive the test booklet, note the **Form** in the bottom left corner, and the Booklet Number printed in black at the top right corner. Fill in the corresponding circles on the answer. And then stop.
3. Do not open the test booklet until we announce.

During the Exam

During the exam:

1. Read all directions carefully.
2. Pay attention to the item numbers on sides 1 and 2 of the answer sheet before you mark a response.
3. **You may write in the test booklet but only responses on the answer sheet will be scored.**
4. We cannot answer any questions during the test. Just do your best.
5. You may want to flip between parts A and B to make best use of time.
6. **You do not need to wait to start Part B.** Follow the pages when you see the words *Go On* until you reach **STOP** at the bottom of page 10.
7. You are graded based on completion and effort.
8. **Place your answer sheet into the test booklet, close it, and hand it in when done,**